

Lukas Helmig

+49 171 623 9717 | lukas.helmig@t-online.de | github.com/lhelmig

PROFESSIONAL SUMMARY

Highly motivated PhD student with strong theoretical foundations in physics and mathematics, and hands-on expertise in numerical simulations and data analysis. Open-minded, curious and quick to thrive in international, fast-paced environments. Strong problem-solving skills with a natural hacker mindset and a passion for turning complex challenges into scalable solutions. Resilient and composed under pressure.

EDUCATION

PhD in Physics Osnabrück University	2028 Osnabrück
Master of Science in Physics University of Cologne	2025 Cologne
Erasmus Exchange L'École normale supérieure	2024 Paris
Bachelor of Science in Physics Osnabrück University	2021 Osnabrück

WORK EXPERIENCE

Research Assistant FZ Jülich GmbH	Oct 2024 – June 2025 Jülich
• Researched Restricted Boltzmann Machines and Hopfield Networks • Developed numerical simulations in Julia and C++ to test scaling laws • Confirmed a theoretical prediction for Hopfield Networks with my simulations	
Research Assistant Osnabrück University	Oct 2021 – Aug 2023 Osnabrück
• Designed custom components using Autodesk Inventor to test out optoelectronic sensors • Sourced parts and hand-Assembled a VORON2.4 3D printer from individual parts • Tweaked custom Klipper firmware for real-time precision control of the 3D printer	
Teaching Assistant Osnabrück University	Oct 2019 – Sept 2021 Osnabrück
• Held tutorial lectures and graded exams • Gave 1-on-1 coaching sessions for atomic physics and optics	

PROJECTS

WanderMark Python, QGIS, OSM/MapLibre	Jan 2026 – Sept 2026
• Developed a personalized map-generation platform for visualizing individual travel adventures • Designed a workflow for GPX route and geotagged photo processing, combining geographic data with user-selected map styles and visual elements • Developed algorithms for automatic photo placement and map layout optimization • Gained hands-on experience with agentic coding environments for AI-assisted software development	

SKILLS & INTERESTS

Technology | Python, C/C++, Java, Mathematica, Pandas, NumPy, Matplotlib
Languages | German (native), English (C1), French (A2), Spanish (A2)
Interests | Design, Pen & Paper, Board games, 3D-Printing, Tennis, Fitness